Recollection G group, $\mathrm{Aut}(G) := \{\text{automorphisms } G \xrightarrow{\sim} G\}$, $\mathrm{Ad}_g := x \mapsto gxg^{-1} \in \mathrm{Aut}(G)$.

Note that $\mathrm{Ad}_g = \mathrm{id}_G \iff gxg^{-1} = x, \forall x \iff gx = xg \iff g \in Z_G$: center of G, subgroup.

§ Cyclic groups

Definition G group. We say G is cyclic if $\exists \sigma \in G, G = \langle \sigma \rangle$, i.e. $\forall g \in G, \exists k \in \mathbb{Z}, g = \sigma^k$.

Example $\forall n \in \mathbb{Z}, (\mathbb{Z}/n\mathbb{Z}, +)$ is cyclic, $\because$ can take $\sigma = 1 + n\mathbb{Z}$.

Want to classify all cyclic groups, up to isomorphism.

Proposition Let $G = \langle \sigma \rangle$ be cyclic.
- If $|G| = \infty$ then $\mathbb{Z} \xrightarrow{\sim} G$ via $n \mapsto \sigma^n$.
- If $n := |G| < \infty$, then $\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} G$ via $k + n\mathbb{Z} \mapsto \sigma^k$.

Proof Can define the homomorphism (surjective) $\mathbb{Z} \twoheadrightarrow G, k \mapsto \sigma^k$.
- If it is injective, we get $\xrightarrow{\sim}$, $|G| = |\mathbb{Z}| = \infty$.
- If not injective, namely $i < j$ s.t. $\sigma^i = \sigma^j \implies \sigma^{j-i} = 1$. Can define $n := \min\{k \geq 1 \mid \sigma^k = 1\}$.

$\forall k \in \mathbb{Z}, \sigma^k$ depends only on $k \bmod n$, $\because \sigma^n = 1$.
$\implies$ Get $\mathbb{Z}/n\mathbb{Z} \twoheadrightarrow G, k + n\mathbb{Z} \mapsto \sigma^k$ still a homomorphism.

Injectivity: suppose $\sigma^i = \sigma^j, 0 \leq i \leq j < n$. Then $\sigma^{j-i} = 1, 0 \leq j - i < n$.
$\implies$ must have $i = j$ by the minimality of n.

Hence $\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} G, |G| = n$. $\square$

As conclusion, cyclic groups are (up to $\simeq$) $\begin{cases} \mathbb{Z} & \text{(infinite)} \\ \mathbb{Z}/n\mathbb{Z},\ n \geq 1 & \text{(finite)} \end{cases}$

$n \neq m \implies \mathbb{Z}/n\mathbb{Z} \not\simeq \mathbb{Z}/m\mathbb{Z}$ by counting elements.

In fact, commutative finite groups are only slightly more complex than cyclic groups. We have the following conclusions:

***Lemma** Let G commutative finite group.
$\forall a, b \in G, \mathrm{ord}(a) = m, \mathrm{ord}(b) = n, \gcd(m, n) = 1 \implies \mathrm{ord}(ab) = mn$.

Proof Say $t := \mathrm{ord}(ab)$. $(ab)^{mn} = (a^m)^n (b^n)^m = 1 \cdot 1 = 1 \implies t \mid mn$.
Also $(ab)^t = 1 \implies a^t = (b^t)^{-1} \implies a^{tn} = (b^{nt})^{-1} = 1 \implies m \mid tn \implies m \mid t$.
Similarly $n \mid t \implies mn \mid t$. $\square$

***Lemma** G commutative finite group, $\exists a \in G$ s.t. $\forall b \in G, \mathrm{ord}(b) \mid \mathrm{ord}(a)$.

Proof Take a s.t. $\forall b \in G, \mathrm{ord}(b) \leq \mathrm{ord}(a)$. To show: $\mathrm{ord}(b) \mid \mathrm{ord}(a)$.

Prove by contradiction: if $\mathrm{ord}(b) \nmid \mathrm{ord}(a)$, then $\exists p$ prime, s.t. $p^s \parallel \mathrm{ord}(a), p^t \parallel \mathrm{ord}(b), s < t$.

Assume $\mathrm{ord}(a) = up^s, \mathrm{ord}(b) = vp^t$, then $\mathrm{ord}(a^{p^s}) = u, \mathrm{ord}(b^v) = p^t$,
$\overset{\text{Lem}}{\implies} \mathrm{ord}(a^{p^s} b^v) = up^t > up^s = \mathrm{ord}(a)$. Contradiction! $\square$

***Proposition** Let G commutative finite group. Then

$$G \text{ cyclic} \iff \forall m \in \mathbb{Z}_{\geq 1}, |\{x \in G \mid x^m = 1\}| \leq m.$$

Proof ($\Longrightarrow$) Let $G = \langle g \rangle, n := |G|$. For $x = g^t, x^m = 1 \implies x^{\gcd(m,n)} = 1$ since $\exists u, v \in \mathbb{Z}$ s.t. $um + vn = \gcd(m, n)$. Hence $g^{t \gcd(m,n)} = 1 \implies n \mid t \cdot \gcd(m, n) \implies \dfrac{n}{\gcd(m, n)} \mid t$.
Therefore $x \in \langle g^{\frac{n}{\gcd(m,n)}} \rangle$, while $\left|\langle g^{\frac{n}{\gcd(m,n)}} \rangle\right| = \gcd(m, n) \leq m$.

($\Longleftarrow$) From Lemma we can take $a \in G$ s.t. $\forall b \in G, \operatorname{ord}(b) \mid \operatorname{ord}(a) =: m$. Hence $\forall b \in G, b^m = 1 \Longrightarrow |\{x \in G \mid x^m = 1\}| = n \leq m$. On the other side $m = \operatorname{ord}(a) \mid n \Longrightarrow m = n$. $\square$

§ Cosets in a group

Let G group, $H \subset G$ subgroup (sometimes denoted as $\leqslant$).

$\forall x, y \in G$, define $x \underset{\text{left}}{\sim} y$ if $\exists h \in H$ s.t. $x = hy$. The right version is similar.

This is a equivalence relation on G :

- $x \underset{\text{left}}{\sim} x, \because x = 1_H \cdot x,$
- $\left(x \underset{\text{left}}{\sim} y \Longrightarrow y \underset{\text{left}}{\sim} x\right), \because x = hy \Longrightarrow y = h^{-1}x,$
- $\left(x \underset{\text{left}}{\sim} y, y \underset{\text{left}}{\sim} z \Longrightarrow x \underset{\text{left}}{\sim} z\right), \because x = hy, y = h'z \Longrightarrow x = (hh')z.$

Recall X set, $\sim$ equivalence relation on X, then

$$X/_{\sim} := \{\sim \text{ equivalence classes in } X\}$$

where $X = \bigsqcup_{C \in X/_{\sim}} C$, with quotient map $X \to X/_{\sim}$.

Now: $\forall g \in G$, set $Hg := \{hg \mid h \in H\}, gH := \{gh \mid h \in H\}$. They are the left (resp. right) equivalence class of g, and called the right (resp. left) coset of g.

The corresponding quotient sets are

$$_H\backslash G := G/_{\underset{\text{left}}{\sim}} = \{\text{right cosets } Hg \mid g \in G\},$$
$$G/_H := G/_{\underset{\text{right}}{\sim}} = \{\text{left cosets } gH \mid g \in G\}.$$

So $G = \bigsqcup_{Hg \in {_H\backslash G}} Hg = \bigsqcup_{gH \in G/_H} gH.$

Lemma $\forall$ subset $S \subset G, S^{-1} := \{s^{-1} \mid s \in S\} \subset G$
$\Longrightarrow (Hg)^{-1} = g^{-1}H^{-1} = g^{-1}H, (gH)^{-1} = Hg^{-1}.$

Proposition $_H\backslash G \longleftrightarrow G/_H$ via

$$Hg \mapsto g^{-1}H = (Hg)^{-1},$$
$$(gH)^{-1} = Hg^{-1} \mapsfrom gH.$$

Proof Both directions are given by $S \mapsto S^{-1}$, and $\left(S^{-1}\right)^{-1} = S$. $\square$

Definition $(G : H) :=$ the common cardinality of $_H\backslash G, G/_H$, called the **index** of $H \subset G$.

Theorem (Lagrange) $|G| = |G| \cdot (G : H)$. (multiplication of cardinal numbers)

Proof $\forall$ coset $C \in {_H\backslash G}$, pick $x_c \in C$, i.e. $Hx_c = C$. Consider

$$H \times {_H\backslash G} \to G,$$
$$(h, C) \mapsto hx_c,$$

Surjective: $\forall g \in G, \exists C$ s.t. $g \in C = Hx_c \Longrightarrow g \in \text{im}$.

Injective: $hx_c = h'x_{c'} \Longrightarrow C \cap C' \neq \emptyset \Longrightarrow C = C', x_c = x_{c'}$. Cancellation law gives $h = h'$.

Hence $|G| = |H| \cdot |{_H\backslash G}| = |H| \cdot (G : H)$.

Another way: $G = \bigsqcup_{C \in {_H\backslash G}} C$; for each C we have $|C| = |H|$ since cancellation law $\Longrightarrow Hg \longleftrightarrow H$.
This also gives $|G| = |H| \cdot |{_H\backslash G}|$. $\square$

Corollary G finite, then $|H|$ divides $|G|$ forall subgroups $H \subset G$.

Corollary $H, K \subset G$ subgroups, finite. $|H|, |K|$ coprime $\Longrightarrow H \cap K = \{1\}$.

Proof $H \cap K$ is a subgroup of both H and K, hence $|H \cap K|$ divides $|H|, |K| \Longrightarrow |H \cap K| = 1$. $\Longrightarrow H \cap K = \{1\}$. $\square$

Definition G group, $\sigma \in G$. **order** / 阶 of σ : $\operatorname{ord}(\sigma) = |\langle \sigma \rangle|$.

Recall that $\operatorname{ord}(\sigma) = \min\{k \geq 1 \mid \sigma^k = 1\}$.

Moreover:

$$\sigma^k = \sigma^l \Longleftrightarrow \operatorname{ord}(\sigma) \mid |k - l|,$$
$$\sigma^k = 1 \Longleftrightarrow \operatorname{ord}(\sigma) \mid k.$$

i.e. $\operatorname{ord}(\sigma) =$ "the minimal period of $\sigma^0, \sigma^1, \sigma^2 \ldots$"

Corollary G finite, $\sigma \in G$, then $\operatorname{ord}(\sigma) \mid |G|$, i.e. $\forall g \in G, g^{|G|} = 1$.

Corollary If $p := |G|$ is a prime number, then G is cyclic.

Proof Note that G : any group, $\sigma \in G, \operatorname{ord}(\sigma) = 1 \Longleftrightarrow \sigma = 1$.

Pick $\sigma \in G, \sigma \neq 1$.

$$1 \neq \operatorname{ord}(\sigma) \mid p \Longrightarrow \operatorname{ord}(\sigma) = p \Longrightarrow |\langle \sigma \rangle| = |G|.$$

Another property of indicies: $K \subset H \subset G$ subgroups, then $(G : K) = (G : H)(H : K)$.

Proof $G = \bigsqcup_{C \in {}_H\backslash G} Hx_c$, pick $x_c \in C$ for all C; $H = \bigsqcup_{D \in {}_K\backslash H} Ky_d$, pick $y_d \in D$ for all D.

$$\begin{aligned} G = \bigsqcup_{C} Hx_c &= \bigsqcup_{C,D} Ky_d x_c \\ \implies {}_K\backslash\, G &\longleftrightarrow \{(C,D) \mid C \in {}_H\backslash\, G, D \in {}_K\backslash\, H\} \\ \implies (G:K) &= (G:H)(H:K). \end{aligned}$$

□

§ Groups in action

In the earlier examples, groups "act" on a set or other structures.

Let X : non-empty set, $\mathrm{S}_x := \{\text{permutations on } X\}$ acts on X, and S_n acts on $\{1, 2, ..., n\}$.

$\forall F$-vs. $V, \mathrm{GL}(V) = \{\text{automorphisms } V \xrightarrow{\sim} V\}$ acts on V.

G group, $\mathrm{Aut}(G)$ acts on G.

Definition G group, X set. A **left G-action on X** means a map $a : G \times X \to X$ s.t.

- $a(1, x) = x$,
- $a(g, a(h, x)) = a(gh, x)$.

Write $a(g, x) =$ "gx". The properties above becomes $1 \cdot x = x, g(hx) = (gh)x$.

Right G-action on X means $a' : X \times G \to X$ s.t. $x \cdot 1 = x, (xg)h = x(gh)$ using similar notations.

For convenience, by default action = left action.

Definition G : any group, define its **opposite group** $G^{\mathrm{op}} := (G, \odot)$ s.t. $g \odot g' = g'g$.
$\implies G^{\mathrm{op}}$ is a group, in which $g^{-1}, 1$ are the same as those in G.

$(G^{\mathrm{op}})^{\mathrm{op}} = G$, and $G^{\mathrm{op}} = G \iff G$ is abelian.

Moreover $G \xrightarrow{\sim} G^{\mathrm{op}}$ via $g \mapsto g^{-1}$.

Left G-action on X = Right G^{op}-action on X. Given $a : G \times X \to X$, can define $a' : X \times G \to X$, $a'(x, g) = a(g, x)$.

Notation $G \curvearrowright X$ or $X \curvearrowleft G$.

Definition Let $G \curvearrowright X, x \in X$.

- The **G-orbit** of $x = Gx := \{gx \mid g \in G\} \subset X$.
- **Stabilizer** of $x = \mathrm{Stab}_G(x) := \{g \in G \mid gx = x\} \subset G$ subgroup.

$\forall x, y \in X$, write $x \underset{G}{\sim} y$ if $\exists g \in G, x = yg$. This is a equivalence relation on X.

- $\underset{G}{\sim}$ equivalence classes $=$ G-orbits Gx in X.

  ${}_G\backslash\, X := X/_{\underset{G}{\sim}} = \{\text{all } G\text{-orbits } Gx\}$.

Next time: Decomposition into orbits.